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manchodu@Srini:~$ cd OSLAB
manchodu@Srini:~/OSLAB$ nano server.c
manchodu@Srini:~/OSLAB$ nano client.c
manchodu@Srini:~/OSLAB$ gcc server.c -o server
manchodu@Srini:~/OSLAB$ gcc client.c -o client
manchodu@Srini:~/OSLAB$ ./server
Server started...
Waiting for client message...
./client
|
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manchodu@Srini:~/OSLAB$ cat server.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define FIFO1 "client_to_server"
#define FIFO2 "server_to_client"

int main()
{
    char buffer[100];

    // Create FIFOs
    mkfifo(FIFO1, 0666);
    mkfifo(FIFO2, 0666);

    printf("Server started...\n");
    printf("Waiting for client message...\n");

    while (1)
    {
        // Open FIFO for reading
        int fd1 = open(FIFO1, O_RDONLY);

        // Read message from client
        read(fd1, buffer, sizeof(buffer));
        close(fd1);

        printf("Client: %s\n", buffer);

        // Process message
        if (strcmp(buffer, "hello\n") == 0 ||
            strcmp(buffer, "hello") == 0)
        {
            strcpy(buffer, "Hello Client!");
        }
        else if (strcmp(buffer, "bye\n") == 0 ||
                 strcmp(buffer, "bye") == 0)
        {
            strcpy(buffer, "Goodbye Client!");
        }
        else
    }
}
```

```

}
manchodu@Srini:~/OSLAB$ cat client.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>

#define FIF01 "client_to_server"
#define FIF02 "server_to_client"

int main()
{
    char message[100];
    char response[100];

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);

    // Open FIFO for writing
    int fd1 = open(FIF01, O_WRONLY);

    // Send message
    write(fd1, message, strlen(message) + 1);
    close(fd1);

    // Open FIFO for reading
    int fd2 = open(FIF02, O_RDONLY);

    // Receive server response
    read(fd2, response, sizeof(response));
    close(fd2);

    printf("Server response: %s\n", response);

    return 0;
}

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manchodu@Srini:~/OSLAB$ gcc server.c -o server
manchodu@Srini:~/OSLAB$ gcc client.c -o client
manchodu@Srini:~/OSLAB$ ./server
Server started...
Waiting for client message...
./client
Client: hello

Server: Hello Client!
Waiting for next client...
Client: Hii ra ela unnav

Server: Hii ra ela unnav
[Processed by Server]
Waiting for next client...
|
```

```
manchodu@Srini:~$ cd OSLAB
manchodu@Srini:~/OSLAB$ ./client
Enter message: hello
Server response: Hello Client!
manchodu@Srini:~/OSLAB$ ./client
Enter message: Hii ra ela unnav
Server response: Hii ra ela unnav
[Processed by Server]
manchodu@Srini:~/OSLAB$ cat server.c
```